Essentials of Genetics

Lecture No. 01

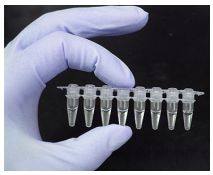
POLYMERASE CHAIN REACTION

PCR: A technique widely used in Molecular Biology and Biotechnology. Its name is from one of its key component - DNA polymerase. As PCR progresses, DNA generated is itself used as template for replication.

PCR- A CHAIN REACTION: This sets in motion of a chain reaction in which the DNA template is exponentially amplified.

HEAT STABLE DNA POLYMERASE: During PCR, a heat-stable DNA polymerase, such as Taq polymerase - an enzyme derived from the bacterium Thermus aquaticus. **CONCLUSION:** PCR can be performed to amplify DNA. It can be extensively modified to perform a wide array of genetic manipulations.

MICROFUGE TUBES: These are small cylindrical plastic tubes with conical bottoms.



THERMOCYCLER: The thermocycler works on the principle of Peltier effect. which raises and lowers the temperature of the block in a pre-programmed manner.



DNA POLYMERASES: Taq polymerase, Pfu polymerase, Vent polymerase

Lecture No. 02

STEPS OF PCR

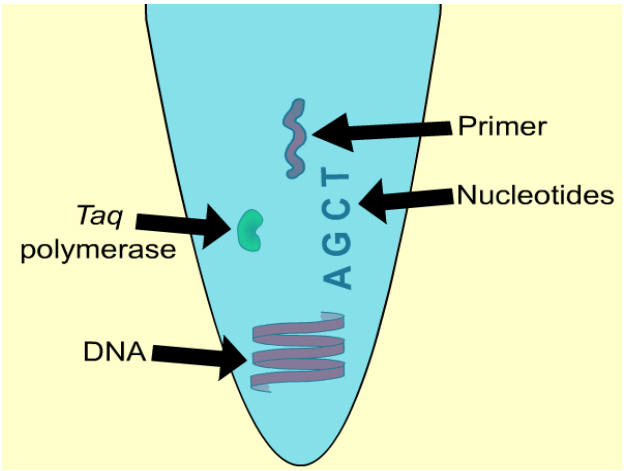
THREE STEPS: Denaturation, Annealing, Extension

INITIALIZATION STEP: This step consists of heating the reaction to a temperature of 93°C - 96°C, which is held for 1-10 minutes.

DENATURATION: First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20-45 seconds. It causes melting of DNA template yielding single strands of DNA.

ANNEALING: The reaction temperature is lowered to 40-65°C for 20-40 seconds. Annealing of the primers to the single-stranded DNA template.

EXTENSION: The temperature at this step depends on the type of DNA polymerase used.Taq polymerase has its optimum activity temperature at 70-80°C. Commonly



72°C is used.

EXTENSION: At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's.

FINAL ELONGATION: Single step is performed at a temperature of 70-74°C for 5-15 minutes after the last PCR cycle to ensure that any remaining single-stranded DNA is fully extended.

FINAL HOLD TEMPERATURE: This step at 4-15°C for an indefinite time may be employed for short-term storage of the reaction.

Lecture No. 03

INGREDIENTS OF PCR

MAJOR INGREDIENTS: Microfuge tube, Thermal cycler, DNA template, Primers, Buffer, MgCl2, Distilled H2o, Deoxynucleotide triphosphates, DNA polymerase

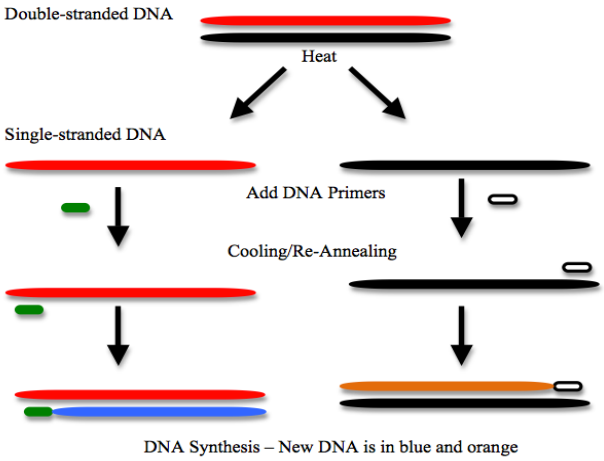
TEMPLATE CAN BE DNA OR RNA: Template DNA, RNA in case of reverse transcriptase

Lecture No. 04

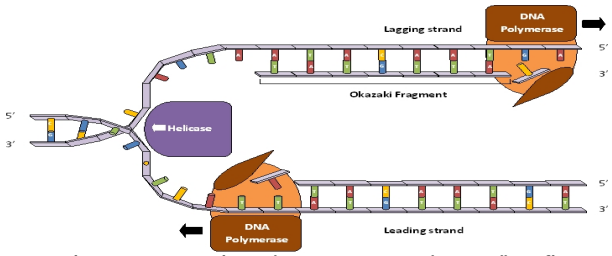
PRIMERS FOR PCR

PRIMERS: Primers are single-stranded 18–30 bp long DNA fragments, Complementary to sequences flanking the region to be amplified.





DNA Synthesis – New DNA is in blue and orange



Taq: *Thermus aquaticus* (most commonly used), *Tfl*: *T. flavus, Tth*: *T. thermophilus*, *Tli*: *Thermococcus litoralis & Pfu*: *Pyrococcus furiosus* (fidelity)

DNA POLYMERASES: Taq polymerase, Pfu polymerase, Vent polymerase

Lecture No. 06

PRIMERS CAN BE SPECIFIC OR RANDOM: Primers determine the specificity of the PCR reaction. Distance between the primers binding sites will determine the size of PCR product.

FEATURES OF PRIMERS: Types of primers-random or specific, Primer length, Annealing temperature, Specificity, Nucleotide composition

PRIMERS:

- Avoid inter-strand homologies
- Avoid intra-strand homologies
- T_m of forward primer = T_m of reverse primer
- G/C content of 20–80%; avoid longer than GGGG
- Product size (100–700 bp)
- Target specificity

STANDARD PCR REACTION

0.25 mM each primer_0.2 mM each dATP, dCTP, dGTP, dTTP_50 mM KCl_10 mM Tris_1.5 mM $MgCl_2$_2.5 units polymerase_10^2 - 10^5 copies of template_**50 ul reaction volume**

PCR TEMPERATURES

- Denaturation temperature:Reduces double stranded molecules to single stranded, 90–96°C, 20-45 seconds
- Annealing temperatureP: Controls specificity of hybridization, 40–68°C, 20-30 seconds
- Extension temperature: Optimized for individual polymerases, 70–75°C, 30-45 seconds

TEMPERATURES

FORMULA FOR CALCULATING MELTING TEMPERATURE OF PRIMERS: Tm = 4(G+C) + 2(A+T)

Lecture No. 05

DNA POLYMERASES

POLYMERASES:

Polymerase	Extension Rate (nt/sec)	Source
Taq pol	75	*T. aquaticus*
Amplitaq (Stoffel fragment)	>50	*T. aquaticus*
Vent	>80	*Thermococcus litoralis*
Pfu	60	*Pyrococcus furiosus*
Tth (RT activity)	>33	*T. thermophilus*

Temp	For	Comments
94-60-72	Perfect, long primers	Higher temp can be used; maximum annealling temp
94-55-72	Good or perfectly matched primers between 19-24 nt	Standard conditions
94-50-72	Adequate primers	Allows 1-3 mismatches/20 nt
94-48-68	Poorly matched primers	Allows 4-5 mismatches/20 nt
94-45-65	Unknown match, likely poor	Primers of questionable quality

Amplification takes place as the reaction mix is subject to an amplification program and temperatures. The amplification program consists of a series of 20–50 PCR cycles.

Lecture No. 07

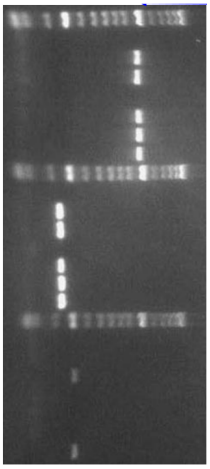
INTERPRETATION OF RESULTS

The PCR product should be of the expected size. Misprimes may occur due to non-specific hybridization of primers.

PRIMERS DIMERS MAY REDUCE AMPLIFICATION: Primer dimers may occur due to hybridization of primers to each other.

BLANK REACTION: Controls for contamination contains all reagents except DNA template.

NEGATIVE CONTROL: Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.



CONCLUSION: The PCR product should be of the expected size.

Lecture No. 08

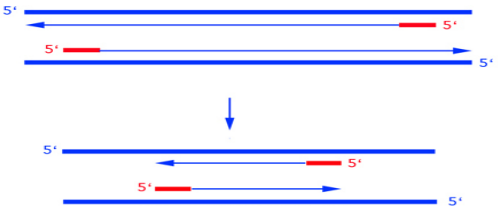
TYPES OF PCR

- **TYPES:** Nested PCR, Multiplex PCR, Touchdown PCR, Sequence-specific PCR, Reverse-transcriptase PCR, Long-range PCR, Whole-genome amplification, RAPD PCR (AP-PCR), Quantitative real-time PCR, Long-range PCR, Whole-genome amplification, RAPD PCR (AP-PCR), Quantitative real-time PCR

Lecture No. 09

NESTED PCR

Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR. Second pair of primers - nested primers (as they lie / are nested within the first fragment).



2ND PCR PRODUCT IS SHORTER THAN FIRST: Second pair of primers bind inside the first PCR product fragment to allow amplification of a second PCR product.

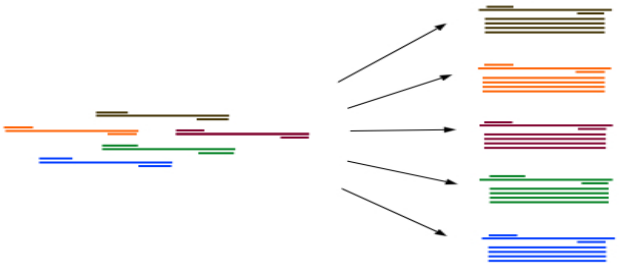
2ND PCR PRODUCT IS SPECIFIC: increases the specificity of DNA amplification, by reducing background due to non-specific amplification of DNA.

ADVANTAGE: Very low probability of nonspecific amplification.

Lecture No. 10

MULTIPLEX PCR

Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. Multiplex-PCR consists of multiple primer sets within a single PCR mixture to produce amplicons of varying sizes that are specific to different DNA sequences.



SINGLE REACTION, MANY AMPLICONS: By targeting multiple sequences at once, additional information may be gained from a single test run that otherwise would require several times the reagents and more time to perform.

Annealing temperatures for each of the primer sets must be optimized to work correctly within a single reaction.

AMPLICON SIZE SHOULD BE DIFFERENT OR BE LABELED: Amplicon sizes should be different enough to form distinct bands when visualized by gel electrophoresis.

LABELED PRIMERS: If amplicon sizes overlap, the different amplicons may be differentiated and visualised using primers that have been dyed with different color fluorescent dyes.

PRIMERS PAIRS BE OPTIMIZED: The primer design for all primers pairs has to be optimized so that all primer pairs can work at the same annealing temperature during PCR.

MULTIPLEX PCR: The use of multiple, unique primer sets within a single PCR reaction to produce amplicons

of varying sizes specific to different DNA sequences.

Lecture No. 11

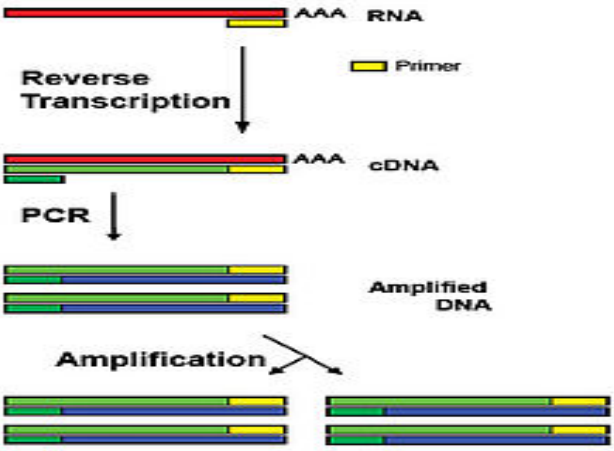
REVERSE TRANSCRIPTASE PCR

Based on the process of reverse transcription, which reverse transcribes RNA into DNA.

First step of RT-PCR - first strand reaction. Synthesis of cDNA using oligo dT primers (37°C) one hour. Second strand reaction - digestion of cDNA:RNA hybrid (RNaseH)-, Standard PCR with DNA oligo primers.



Quantitative real-time PCR is often confusingly known as RT-PCR (Real Time PCR) or RQ-PCR. QRT-PCR or RTQ-PCR are more appropriate contractions. QRT-PCR methods use fluorescent dyes, such as Sybr Green, or



fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time. Q-PCR is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample.

Lecture No. 13

HOT START PCR

This is a technique that reduces non-specific amplification during the initial set up stages of the PCR. The technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C). Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature. This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step.

ENZYME REVERSE TRANSCRIPTASE: The PCR is preceded by a reaction using reverse transcriptase to convert RNA to cDNA

REVERSE TRANSCRIPTASE PCR: RT-PCR is widely used in expression profiling, to determine the expression of a gene. To identify the sequence of an RNA transcript.

Lecture No. 12

REAL TIME PCR

Real Time PCR is a technique in which fluoroprobes bind to specific target regions of amplicons to produce fluorescence during PCR. Real Time PCR is used to measure the quantity of a PCR product. The fluorescence, measured in Real Time, is detected in a PCR cycler with an inbuilt filter flurometer. It is the method of choice to quantitatively measure starting amounts of DNA, cDNA or RNA.

DNA Polymerase- Eubacterial type I DNA polymerase, Pfu. These thermophilic DNA polymerases show a very small polymerase activity at room temperature .

Lecture No. 14

ASYMMETRIC PCR

Asymmetric PCR is used to amplify one strand of the original DNA more than the other.

It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal.

PCR is carried out as usual, but with a great excess of one primers for the chosen strand.

Due to the slow amplification later in the reaction after the limiting primer has been used up, extra cycles of PCR are required. It finds use in some types of sequencing and hybridization probing where having only one of the two complementary strands is required.

Lecture No. 15

LONG PCR

Extended or longer than standard PCR, meaning over 5 kilobases (frequently over 10 kb). Long PCR is useful only if it is accurate.

MIXTURES OF POLYMERASES: Special mixtures of proficient polymerases along with accurate polymerases such as Pfu are often mixed together.

LONG PCR: Application - to clone large genes not possible with conventional PCR.

Lecture No. 16

ALLELE SPECIFIC PCR

Allele-specific PCR used for identifying of SNPs. It requires prior knowledge of a DNA sequence, including differences between alleles. Uses primers whose 3' ends encompass the SNP. PCR amplification under stringent conditions is much less efficient in the presence of a mismatch between template and primer.

AMPLIFICATION WITH SNP SPECIFIC PRIMER: Successful amplification with an SNP-specific primer signals presence of the specific SNP in a sequence.

ALLELE SPECIFIC PCR: This diagnostic or cloning technique is used to identify or utilize single-nucleotide polymorphisms (SNPs).

Lecture No. 17

COLONY PCR

The screening of bacterial or yeast clones for correct ligation or plasmid products.

METHODOLOGY: The screening of bacterial or yeast clones for correct ligation or plasmid products. Pick a bacterial colony with an autoclaved toothpick, swirl it into Transfer 20 µl of the supernatant into a new microfuge tube. Take 1-2 µl of the supernatant as template in a 25 µl PCR standard PCR reaction.

The screening of bacterial or yeast clones for correct ligation or plasmid products.

Lecture No. 18

IN SITU PCR

In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide. In situ PCR amplification can be performed on fixed tissue or cells . Applies the methodology of hybridization of the nucleic acids. Allows identification of cellular markers Limited to detection of non-genomic material such as RNA.

Lecture No. 19

INVERSE PCR

Inverse PCR uses standard PCR primers oriented in the reverse direction of the usual orientation.

The template for the reverse primers is a restriction fragment that has been self-ligated. Inverse PCR functions to clone sequences flanking a known sequence. Flanking DNA sequences are digested and then ligated to generate circular DNA.

IDENTIFICATION OF SEQUENCES FLANKING TRANSPOSABLE ELEMENTS: Amplification and identification of sequences flanking transposable elements, and the identification of genomic inserts.

INVERSE PCR: A method used to allow PCR when only one internal sequence is known. Especially useful in identifying flanking sequences to various genomic inserts.

Lecture No. 20

AFLP PCR

METHODOLOGY: Genomic DNA is digested with one or more restriction enzymes. tetracutter (MseI) and a hexacutter (EcoRI). Ligation of linkers to all restriction fragments. Pre-selective PCR is performed using primers which match the linkers and restriction site specific sequences. Electrophoretic separation and amplicons on a gel matrix, followed by visualization of the band pattern.

AFLP is a highly sensitive PCR-based method for detecting polymorphisms in DNA. AFLP can be also used for genotyping individuals for a large number of loci.

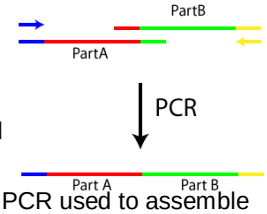
Lecture No. 21

ASSEMBLY PCR

Assembly PCR used to assemble two or more pieces of DNA into one piece.

PRINCIPLE: Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more pieces of DNA into one piece.

METHODOLOGY: It involves an initial PCR with primers that have an overlap and a second PCR using the products as the template that generates the final full-length product.



ASSEMBLY PCR: Assembly PCR used to assemble two or more pieces of DNA into one piece.

Lecture No. 22

SUICIDE PCR

PRINCIPLE: Suicide PCR is typically used in paleogenetics or other studies where avoiding false positives and ensuring the specificity of the amplified fragment is the highest priority. The method prescribes the use of any primer combination only once in a PCR, which should never have been used in any positive control PCR reaction.

SUICIDE PCR: Primers should always target a genomic region never amplified before in the lab using this or any other set of primers.This ensures that no contaminating DNA from previous PCR reactions is present in the lab, which could otherwise generate false positives. Use of any primers combination only once in a PCR.

Lecture No. 23
METHYLATION SPECIFIC PCR

Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA. Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

MSP: MSP used in quantitative PCR provides information about methylation state of a given CpG island.

Lecture No. 24
INTER SEQUENCE SPECIFIC PCR

A method for DNA fingerprinting that uses primers selected from segments repeated throughout a genome to produce a unique fingerprint of amplified product lengths. The use of primers from a commonly repeated segment is called Alu-PCR, and can help to amplify sequences adjacent (or between) these repeats.

PCR: PCR used to produce a unique fingerprints of amplified product lengths.

Lecture No. 25
LIGATION MEDIATED PCR

METHODOLOGY: Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA. PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

PCR: DNA sequencing, Genome walking, DNA footprinting

Lecture No. 26
WHOLE GENOME AMPLIFICATION PCR

Primers can be designed to be 'degenerate'

- able to initiate replication from a large number of target locations. Whole genome amplification (WGA) is a group of procedures that allow amplification to occur at many locations in a genome.

PCR: Whole genomes can be amplified by WGA.